Idaho Technology Authority (ITA)

ENTERPRISE STANDARDS S3000 – NETWORK AND TELECOMMUNICATIONS

Category: S3220 – SECURITY – VIRTUAL PRIVATE NETWORK

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I. DEFINITION

See ITA Guideline G105 (Glossary of Terms) for any definitions. G105-ITA Glossary of Terms

II. RATIONALE

There are many business requirements that justify providing logical access to network resources, independent of physical topology or location. A VPN implementation can be used to fulfill these business requirements while maintaining security standards. Any VPN solution must have an architecture that can be implemented within the enterprise security policy, including passwords, smart cards, token-based products, directory systems (LDAP, etc.), remote access services, and X.509 digital certificates.

III. APPROVED STANDARD(S)

1. IPsec (IKEv2) for site-to-site VPNs.

2. TLS-based remote access VPN configured for TLS 1.2/1.3 with AEAD cipher suites (AES-GCM/ChaCha20-Poly1305). CBC ciphers are not authorized without a firewall exception.

3. SAML 2.0 for SSO and federated identity with MFA enforced at IdP.

4. Zero Trust/SDP/ZTNA patterns for private-app access where feasible.

IV. APPROVED PRODUCT(S)

Refer to approved state contracts for IT Security Products:
https://purchasing.idaho.gov/statewide-contracts/

V. JUSTIFICATION

Products provided in the state contracts are recognized as industry leaders.

VI. TECHNICAL AND IMPLEMENTATION CONSIDERATIONS

Enforce MFA for all VPN access with SSO via SAML 2.0 IdP; configure GlobalProtect SAML Authentication Profile (GP-SAML-Auth-ID). Require TLS 1.2/1.3 AEAD ciphers; IPsec IKEv2 for site-to-site. Default full-tunnel for sensitive workloads; split-tunnel only by documented exception. Log authentication events and forward to SIEM, or log collector; enforce posture checks; apply formal change control.

VII. EMERGING TRENDS AND ARCHITECTURAL DIRECTIONS

Zero Trust Architecture (ZTA), SASE/SSE adoption, TLS modernization (TLS 1.3), federated authentication (SAML/OAuth/OIDC).

VIII. PROCEDURE REFERENCE

Pairs with ITS ISPM v4.0 (AC-17 Remote Access, S.IA-01 Authentication, AU-01/06 Logging, SC-07 Boundary Protection), NIST SP 800-77, NIST SP 800-52 Rev.2, OASIS SAML v2.0. CBC cipher use requires approval.

IX. REVIEW CYCLE

Every 3 years.

X. CONTACT INFORMATION

For more information, contact the ITA Staff at (208) 605-4064.

REVISION HISTORY

05/05/2026 - Updates refine wording, standardize formatting, remove redundant text, and shift definitions to G105 for a more streamlined VPN standard.

07/01/13 – Changed “ITRMC” to “ITA”.

6/16/09 – Deleted Timeline from standard.

3/7/07 – Added additional considerations regarding new or replacement remote access solutions.

9/13/06 – Added SSL as an approved standard. Added F5 FirePass as an approved SSL product. Updated technical considerations and procedural reference.

4/25/05 – Added Cisco VPN products.

8/25/04 – Acknowledged emerging trend of SSL-based VPNs.

Effective Date: October 2, 2001